

HELSINGBORG, Sweden

WMO: 026110

Lat: 56.030N

Lon: 12.765E

Elev: 44

StdP: 100.80

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.8	(c) -7.4	(d) -12.9	(e) 1.2	(f) -8.4	(g) -10.5	(h) 1.5	(i) -6.1	(j) 11.3	(k) 4.6	(l) 10.1	(m) 4.8	(n) 2.6	(o) 40	(p) 0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	26.4	18.8	24.7	18.0	23.0	17.2	20.0	24.3	18.9	22.9	18.0	21.8	3.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.4	21.7	17.4	12.5	20.8	16.5	11.8	19.9	57.3	24.2	53.9	22.7	51.1	21.8	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.0	(b) 7.8	(c) 7.0	(e) -13.3	(f) 28.7	(g) 3.8	(h) 1.6	(i) -16.0	(j) 29.9	(k) -18.3	(l) 30.9	(m) -20.4	(n) 31.8	(o) -23.2	(p) 33.0		
(a) 9.0	(b) 7.8	(c) 7.0	(e) -13.6	(f) 21.6	(g) 3.8	(h) 1.2	(i) -16.4	(j) 22.4	(k) -18.6	(l) 23.1	(m) -20.7	(n) 23.8	(o) -23.4	(p) 24.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	0.5	0.9	3.0	7.5	11.8	15.0	17.3	17.3	13.9	9.2	5.1	2.1	(6)
(7)		DBStd	6.88	4.02	3.68	3.53	3.27	3.23	2.59	2.47	2.36	2.55	3.13	3.32	3.99	(7)
(8)		HDD10.0	1319	295	256	218	88	16	1	0	0	2	51	148	244	(8)
(9)		HDD18.3	3568	553	489	476	324	203	106	51	49	134	283	397	502	(9)
(10)		CDD10.0	837	0	0	1	14	73	149	227	227	119	26	2	0	(10)
(11)		CDD18.3	44	0	0	0	0	2	5	19	17	1	0	0	0	(11)
(12)		CDH23.3	308	0	0	0	1	16	47	152	88	4	0	0	0	(12)
(13)		CDH26.7	35	0	0	0	0	1	4	23	6	0	0	0	0	(13)
(14)	Wind	WSAvg	3.5	3.7	3.9	3.6	3.6	3.4	3.3	3.0	3.1	3.3	3.5	3.7	3.9	(14)
(15)	Precipitation	PrecAvg	723	57	43	39	36	50	74	77	87	70	67	60	63	(15)
(16)		PrecMax	962	129	107	82	69	111	164	210	183	140	140	124	113	(16)
(17)		PrecMin	481	1	5	13	8	19	0	0	17	13	18	12	12	(17)
(18)		PrecStd	103	30	19	19	18	24	39	48	42	33	29	26	25	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	9.1	13.9	20.9	25.5	27.3	29.1	27.5	23.6	17.2	12.7	10.1	(19)
(20)			MCWB	7.9	7.8	8.9	13.7	17.0	18.9	19.7	19.1	17.5	14.0	11.9	9.4	(20)
(21)		2%	DB	7.4	7.7	10.9	17.6	22.6	24.6	26.9	25.7	21.3	15.3	11.5	9.1	(21)
(22)			MCWB	6.6	6.5	7.4	10.8	15.7	17.5	19.3	18.6	16.5	13.4	10.6	8.3	(22)
(23)		5%	DB	6.5	6.7	9.0	15.5	20.2	22.3	25.0	24.0	19.5	14.2	10.4	8.0	(23)
(24)			MCWB	5.7	5.7	6.5	10.3	14.1	16.4	18.3	18.0	15.7	12.6	9.4	7.3	(24)
(25)		10%	DB	5.6	5.7	7.6	13.3	18.0	20.3	23.0	22.2	18.0	13.3	9.4	7.1	(25)
(26)			MCWB	4.8	4.8	5.8	9.1	12.9	15.4	17.5	17.3	14.8	11.9	8.5	6.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	8.1	9.6	14.4	18.5	20.3	21.6	21.1	18.4	14.9	12.2	9.7	(27)
(28)			MCD B	8.7	8.7	12.2	19.5	23.4	24.9	26.6	25.2	21.9	16.1	12.6	10.0	(28)
(29)		2%	WB	6.8	6.8	8.0	12.0	16.5	18.3	20.1	19.7	17.1	13.7	10.7	8.4	(29)
(30)			MCD B	7.3	7.4	10.1	16.3	21.2	22.8	24.8	23.3	20.3	14.9	11.3	8.9	(30)
(31)		5%	WB	5.8	5.8	7.0	10.8	15.0	17.1	19.2	18.7	16.1	12.9	9.6	7.4	(31)
(32)			MCD B	6.3	6.5	8.6	14.5	18.8	21.2	23.4	22.2	18.7	14.0	10.2	7.9	(32)
(33)		10%	WB	4.8	4.8	6.0	9.5	13.5	16.0	18.2	17.9	15.3	12.1	8.7	6.4	(33)
(34)			MCD B	5.5	5.7	7.3	12.5	17.0	19.5	22.1	21.3	17.5	13.1	9.3	7.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.9	4.1	5.9	8.0	8.6	8.1	8.2	7.8	6.9	5.3	3.8	3.6	(35)
(36)			MCDBR	3.7	4.5	8.1	11.5	12.5	12.0	12.6	10.8	9.0	5.8	4.0	3.8	(36)
(37)			MCWBR	3.7	3.7	5.5	6.7	7.0	6.3	6.0	5.2	5.2	4.2	3.7	3.7	(37)
(38)		5% WB	MCD BR	3.6	4.2	7.1	10.1	10.8	10.5	10.5	9.0	8.0	5.5	3.8	3.7	(38)
(39)			MCWBR	3.7	3.7	5.3	6.2	6.7	6.3	5.8	5.2	5.1	4.3	3.7	3.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.308	0.332	0.354	0.387	0.384	0.378	0.400	0.400	0.369	0.353	0.333	0.300	(40)	
(41)		taud	2.274	2.302	2.317	2.285	2.348	2.391	2.347	2.355	2.425	2.399	2.324	2.252	(41)	
(42)		Ebn at Noon	590	726	805	829	856	866	835	809	785	696	555	499	(42)	
(43)		Edh at Noon	54	77	96	114	114	110	113	106	86	70	52	44	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.43	1.02	2.39	4.07	5.28	5.58	5.17	4.16	2.94	1.43	0.54	0.29	(44)	
(45)		RadStd	0.08	0.16	0.26	0.52	0.52	0.46	0.62	0.33	0.31	0.21	0.10	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+101	+21
(47) Regional (64 neighbors)		+0.40	N/A	N/A	N/A	N/A	N/A	N/A	-157	N/A	N/A

Nomenclature: See separate page